

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story Points)

Date	30 June 2026	Name	Nadin Surya S
Project Name	Meeting Room Booking System using ServiceNow App Engine Studio		
Maximum Marks	5 Marks		

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1 - Application Setup, Database & Form Design (20 pts)						
Sprint-1	Application Setup	USN-1	As an admin, I can create the Meeting Room Booking application in App Engine Studio so the project has a working foundation.	3	High	Nadin Surya S
Sprint-1	Application Setup	USN-2	As an admin, I can configure application roles (Admin and User) so access to the app is properly controlled.	2	High	Nadin Surya S
Sprint-1	Database Design	USN-3	As a developer, I can create a custom Meeting Room Booking table so booking data can be stored centrally.	3	High	Nadin Surya S
Sprint-1	Database Design	USN-4	As a developer, I can configure auto-numbering with prefix "MRB" so each booking record gets a unique ID.	2	Medium	Nadin Surya S
Sprint-1	Database Design	USN-5	As a developer, I can add all required fields (employee, room, date, time, purpose, status) to the booking table.	3	High	Nadin Surya S
Sprint-1	Database Design	USN-6	As an admin, I can define choice values for Room Name and Status so users select only valid options.	2	Medium	Nadin Surya S
Sprint-1	Form Design	USN-7	As a user, I can view a booking form with all required fields so I can submit a meeting room request.	3	High	Nadin Surya S
Sprint-1	Form Design	USN-8	As an admin, I can configure mandatory fields on the form so incomplete bookings cannot be submitted.	2	Medium	Nadin Surya S
Sprint-2 - Business Logic & Workflow Automation (20 pts)						
Sprint-2	Validation	USN-9	As a user, the End Time of my booking is validated against the Start Time via Client Script before submission.	3	High	Nadin Surya S
Sprint-2	Validation	USN-10	As a user, I see an error message if End Time is less than or equal to Start Time, so the booking is not accepted.	2	Medium	Nadin Surya S
Sprint-2	Status Automation	USN-11	As a system, a Business Rule automatically sets a new booking's status to "Pending" before it is inserted.	2	High	Nadin Surya S
Sprint-2	Workflow Automation	USN-12	As a developer, I can create a Flow Designer flow that triggers automatically when a new booking record is created.	3	High	Nadin Surya S
Sprint-2	Workflow Automation	USN-13	As a system, the flow updates the booking status automatically as part of the workflow.	2	Medium	Nadin Surya S
Sprint-2	Notification	USN-14	As a user, I receive an email notification once my booking request has been created.	3	High	Nadin Surya S
Sprint-2	Booking Management	USN-15	As an admin, I can approve a pending meeting room booking request.	3	High	Nadin Surya S
Sprint-2	Booking Management	USN-16	As an admin, I can mark an approved booking as "Completed" once the meeting is over.	2	Medium	Nadin Surya S

Sprint-3 - Dashboard & Reporting (20 pts)						
Sprint-3	Dashboard	USN-17	As an admin, I can view a widget showing the total number of bookings made.	2	Medium	Nadin Surya S
Sprint-3	Dashboard	USN-18	As an admin, I can view a chart of bookings grouped by status (Pending / Approved / Completed).	3	High	Nadin Surya S
Sprint-3	Dashboard	USN-19	As an admin, I can view a room-wise booking chart to see which rooms are used most.	3	Medium	Nadin Surya S
Sprint-3	Dashboard	USN-20	As an admin, I can view a booking trend chart showing booking volume over time.	3	Medium	Nadin Surya S
Sprint-3	Dashboard	USN-21	As an admin, I can view a full list of bookings directly on the dashboard.	2	Low	Nadin Surya S
Sprint-3	Reports	USN-22	As an admin, I can generate a Booking Status Report for review and audit.	3	High	Nadin Surya S
Sprint-3	Reports	USN-23	As an admin, I can generate a Room-wise Booking Report to analyze room utilization.	2	Medium	Nadin Surya S
Sprint-3	Reports	USN-24	As an admin, I can generate a Booking Trend Report to study usage patterns over time.	2	Low	Nadin Surya S
Sprint-4 - Testing & Validation (20 pts)						
Sprint-4	Testing	USN-25	As a tester, I can test end-to-end booking creation to confirm the workflow functions correctly.	3	High	Nadin Surya S
Sprint-4	Testing	USN-26	As a tester, I can validate that the Client Script correctly blocks invalid time entries.	3	High	Nadin Surya S
Sprint-4	Testing	USN-27	As a tester, I can verify that the Business Rule sets the default status correctly on every new booking.	2	High	Nadin Surya S
Sprint-4	Testing	USN-28	As a tester, I can verify Flow Designer executes and updates booking status as expected.	3	High	Nadin Surya S
Sprint-4	Testing	USN-29	As a tester, I can confirm that email notifications are delivered for booking events.	2	Medium	Nadin Surya S
Sprint-4	Testing	USN-30	As a tester, I can verify that dashboard widgets reflect accurate, up-to-date booking data.	3	Medium	Nadin Surya S
Sprint-4	Testing	USN-31	As a tester, I can verify that all reports generate correct and complete output.	2	Medium	Nadin Surya S
Sprint-4	Go-Live Review	USN-32	As an admin, I can review the complete booking history before the application goes live.	2	Low	Nadin Surya S

Project Tracker, Velocity & Burndown Chart (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	01 Jun 2026	06 Jun 2026	20	06 Jun 2026
Sprint-2	20	6 Days	08 Jun 2026	13 Jun 2026		
Sprint-3	20	6 Days	15 Jun 2026	20 Jun 2026		
Sprint-4	20	6 Days	22 Jun 2026	27 Jun 2026		

Velocity

For the Meeting Room Booking System, each sprint runs for **6 days** and the team completes **20 story points** per sprint (as demonstrated in Sprint-1). The team's average velocity (AV) per iteration unit – i.e. story points completed per day – is calculated as:

$$AV = \text{velocity} \div \text{sprint duration} = 20 \div 6 = \mathbf{3.33 \text{ story points / day}}$$

This means the team consistently delivers approximately 3.33 story points of work each day, which can be used to forecast completion dates for future sprints and the overall release.

Burndown Chart - Sprint-1

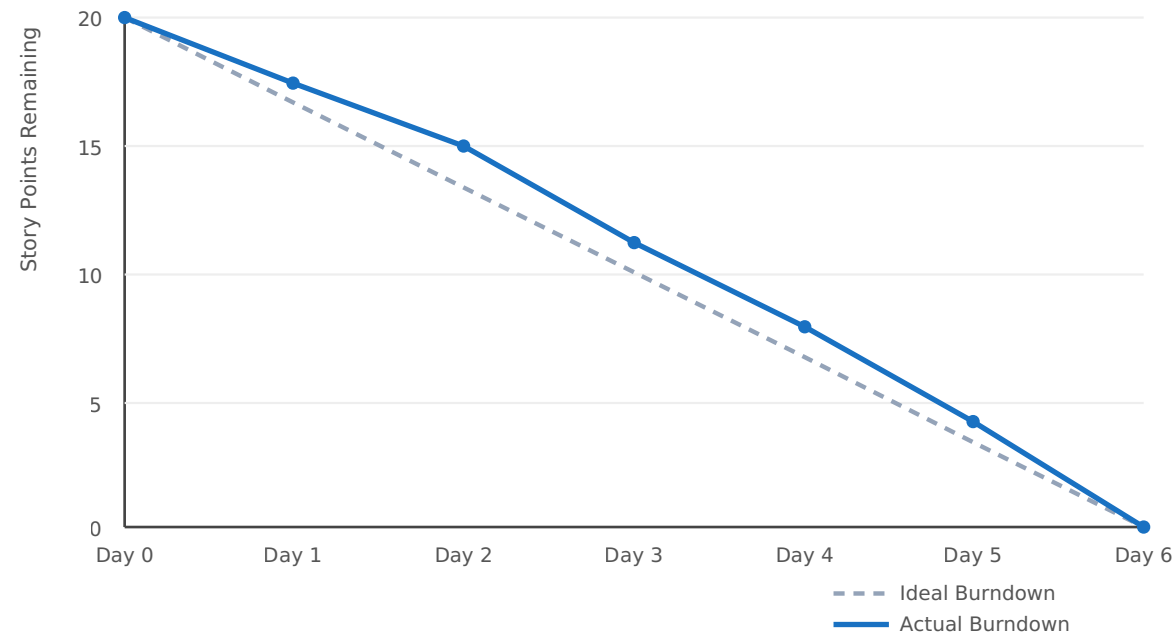


Figure 1: Sprint-1 burndown chart - story points remaining vs. time, comparing ideal and actual progress

A burndown chart is a graphical representation of work remaining versus time, commonly used in Scrum and other agile methodologies to track sprint progress and forecast completion. For Sprint-1 of the Meeting Room Booking System, the actual burndown closely tracked the ideal line and reached zero remaining story points on Day 6, confirming that all 20 planned story points were completed on schedule.

Reference:

<https://www.atlassian.com/agile/project-management>
<https://www.atlassian.com/agile/tutorials/epics>
<https://www.atlassian.com/agile/tutorials/sprints>
<https://www.atlassian.com/agile/project-management/estimation>
<https://www.atlassian.com/agile/tutorials/burndown-charts>
<https://www.visual-paradigm.com/scrum/scrum-burndown-chart/>